

Profile: IoT and Embedded Systems Engineer with hands-on experience in **embedded systems design, firmware development, and IoT solutions development**. Expertly developed firmware, integrated diverse sensors, and implemented **real-time data acquisition and control systems**. Familiar with **Agile development practices** and system validation methodologies for reliable embedded system deployment. **ESP32, Arduino, and Raspberry Pi-based systems**. Skilled in **sensor interfacing, real-time data acquisition, and system integration**. Experienced in building intelligent systems for automation, energy optimization, and security with a focus on reliability and performance.

Core Strengths: Embedded C • Sensor Integration • Communication Protocols • Real-time Systems • System Validation

SKILLS

Programming	C, C++, Embedded C, Python
MCU Platforms	ESP32 (Xtensa), Arduino (AVR), Raspberry Pi (ARM), Jetson Nano
Protocols	UART, I2C, SPI, MQTT, HTTP/REST
Hardware	ADC, PWM, GPIO, Relay Control, Voltage/Current Sensors
Systems	Data Acquisition, Real-time Monitoring, Automation, Edge Processing
Validation	System Testing, Integration Testing, Sensor Calibration, Data Validation
Concepts	Interrupt Handling, Timer-based Execution, Digital Signal Filtering (Basic)
Tools	Arduino IDE, ThingSpeak VS Code, Git, Docker, Proteus, Github Action, Git
Process	SDLC, Agile

EXPERIENCE

AI & Hardware Developer	2016 — 2019
AI ROBOT BONDHU, ICT Ministry Bangladesh	
- Developed and tested embedded firmware for AI-integrated hardware systems. - Performed sensor interfacing, actuator control, and communication integration across multiple modules. - Conducted system-level testing and debugging, identifying integration issues between hardware and software components. - Worked on real-time system behavior validation to ensure stability and performance under varying conditions. - Optimized system performance through iterative testing and debugging of embedded modules.	

PROJECTS

Intelligent Solar Panel System	2024–2025
- Designed and developed a real-time energy monitoring and optimization system using ESP32 and Jetson Nano. - Integrated environmental sensors (BME280, BH1750) via I2C for temperature, humidity, and light-based analysis. - Implemented PWM-based MPPT/PWM charge control logic to optimize energy harvesting and battery charging efficiency. - Developed relay-based protection system for overvoltage, overcurrent, and fault handling mechanisms. - Built data acquisition pipeline for continuous monitoring and logging of voltage, current, and environmental parameters. - Performed sensor calibration, system testing, and performance validation to ensure accurate readings and stable operation. - Conducted energy efficiency analysis and fault detection testing under different environmental conditions.	
Smart Home Ecosystem	2025
- Designed a multi-module IoT system integrating security, wellness monitoring, and automation control. - Implemented MQTT-based communication architecture for real-time data exchange between devices. - Developed interrupt-driven event handling for motion detection and responsive automation. - Integrated multiple sensors and control systems using GPIO and ADC for real-time data acquisition. - Designed automation workflows for dynamic system response based on sensor input. - Performed system integration testing and reliability validation for continuous operation.	
Traffic Management System	2024
- Developed a sensor-based traffic decision system using gyroscope, accelerometer, and magnetometer. - Implemented low-latency firmware algorithms for edge-level decision making.	
Smart Security System	2016
- Designed Arduino-based system integrating GSM module, sonar sensors, and keypad. - Implemented real-time alert system for security monitoring.	
Affordable Prosthetic Hand	2019
- Developed a low-cost prosthetic prototype using servo motors and 3D printing. - Designed mechanical actuation system for basic human motion assistance.	

RESEARCH

- **Design & Implementation of Intelligent Solar Panel System** — IEEE CS BDC Summer Symposium 2025.
- Developed a **smart solar energy system** focusing on monitoring, automation (cleaning and power sharing), and protection from hailstorms.
- Applied **system testing, sensor validation, and data analysis** to evaluate performance.
- Conducted **efficiency testing and fault-handling validation** under different environmental conditions.
- Demonstrated reliable system behavior and stable energy monitoring through structured testing and validation.

EDUCATION

B.Sc. in Computer Science and Engineering
Ahsanullah University of Science and Technology

2021 — 2025

TECHNICAL & LEADERSHIP ACTIVITIES

- Executive Member — AUST Robotics Club (2021–2025): Led technical activities and collaborated on robotics system development
- Executive Member — AUST Programming & Information Club: Participated in problem-solving and development sessions
- Campus Ambassador — ICT Olympiad Bangladesh: Coordinated student engagement and tech awareness programs
- Organizing Committee Member - AUST CSE Carnival 5.0, Annual Picnic, CSE Cricket Tournament, CSE Cultural Fest — Managed logistics, coordination, and team execution for large-scale events

ACHIEVEMENTS

2019	BCSIR National Science Fest Runner-up — Prosthetic Hand Project
2016	BCSIR National Science Fest Champion — Smart Security System
2024	Robomania Runner-up — Robotics Competition
2024	Cefalo Carnival Runner-up — Soccer Bot System
2022-2025	AUST CSE Sports Achievements — AC RED League Football Tournament, Champion; Futsal Championship, Champion; Football, Championship Runner-up ; Cricket Tournament, Runner-up